

S/4HANA 2023 Finance & Controlling Cutover

Architecture Review Document — Material Ledger Focus

Migration Path	ICOST → SAP ECC → SAP S/4HANA 2023
Version	1.1
Focus Area	Material Ledger Cutover Sequencing
Base Scenario	Mid-Year Cutover (Scenario B)
Status	Draft — Architecture Review
Date	May 2026
Classification	Intel Confidential

Table of Contents

- 1. [Executive Summary](#)
- 2. [Cutover Scenario Matrix](#)
- 3. [ICOST Migration Context](#)
- 4. [End-to-End Architecture Overview](#)
- 5. [Material Ledger — What Changes in S/4HANA 2023](#)
- 6. [Material Extension — Per-Plant Requirements](#)
- 7. [Material Ledger Cutover — The Core Cycle](#)
- 8. [WIP Balances — Design Decision](#)
- 9. [Master Cutover Sequence](#)

1. Executive Summary

This document consolidates architecture decisions, data flows, configuration dependencies, and cutover sequencing for a migration from an ICOST-based costing engine (feeding GL postings to SAP ECC) to SAP S/4HANA 2023 with:

- **Account-Based CO-PA / Margin Analysis** (Universal Journal — ACDOCA)
- **Material Ledger with actual costing** (mandatory in S/4HANA)
- **COGS Split** to cost components
- **SAC for planning** (writing to ACDOCP)
- **Mid-year cutover** assumed (most complex scenario)

The primary refinement in this version is the Material Ledger cutover cycle — upload timing, per-plant/per-site sequencing, material extension requirements, and the initialization sequence that must precede any transactional go-live.

2. Cutover Scenario Matrix

Three Cutover Scenarios to Consider

Scenario	Description	Complexity	Recommended?
A — Year-End Cutover	Go-live on first day of new fiscal year	Low	Yes — if timeline allows
B — Mid-Year Cutover	Go-live during fiscal year	High	Only if business-mandated
C — Mid-Period Cutover	Go-live mid-period (not period end)	Very High	Avoid — closing risk extreme

This document designs for **Scenario B (Mid-Year)** as the base case — it subsumes Scenario A and covers Scenario C with additional controls.

3. ICOST Migration Context

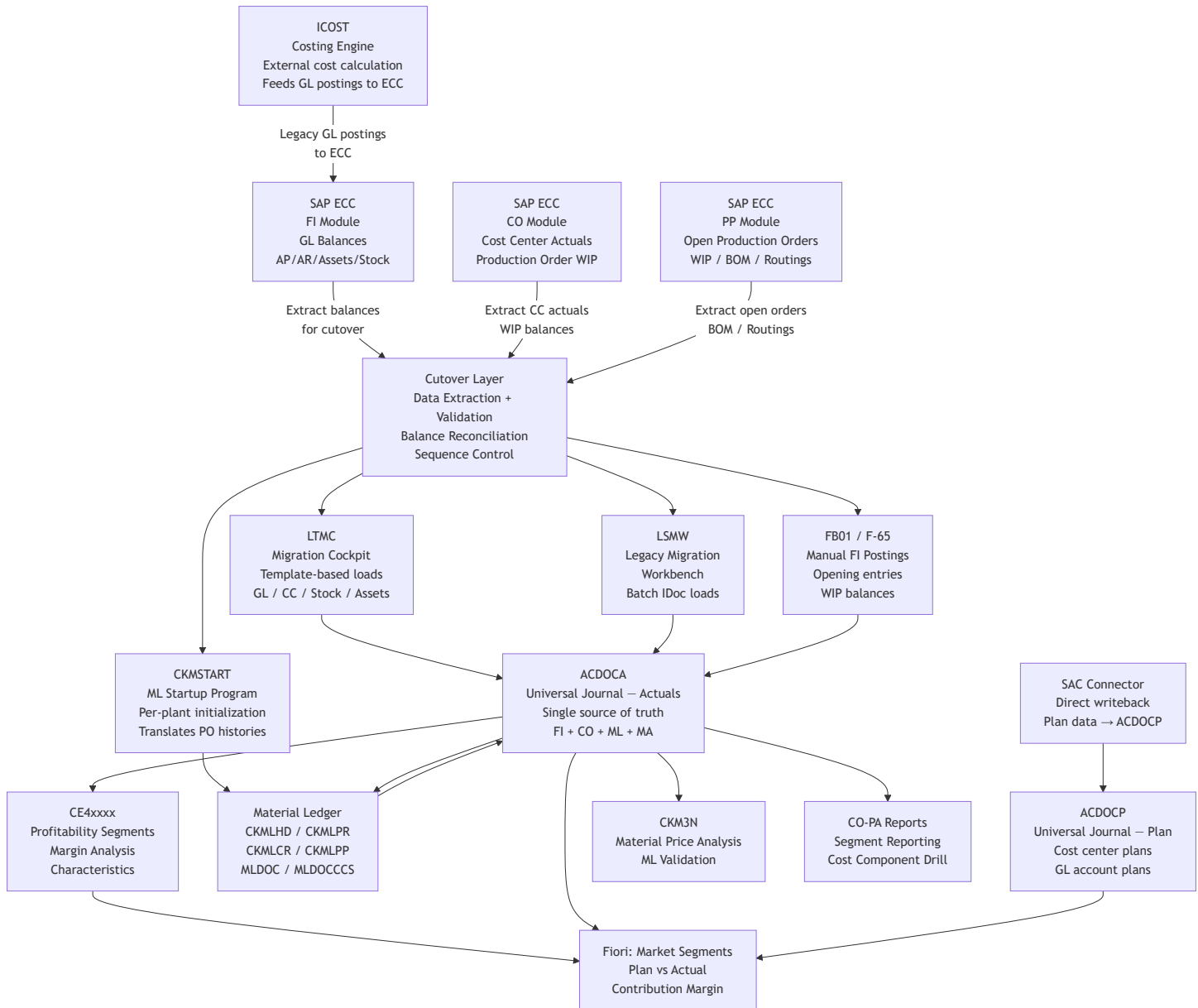
What ICOST Does vs. What S/4HANA Replaces It With

Capability	ICOST + ECC	S/4HANA 2023 Native
Cost calculation engine	External ICOST engine	CO-PC — standard cost estimates (CK11N)
GL posting generation	ICOST → ECC via interface	SAP accounting interface — real-time to ACDOCA
Cost component split	ICOST-defined components	Cost component structure in CO-PC
Material valuation	MAP or standard in ECC	Material Ledger mandatory — actual costing
Variance calculation	ICOST-driven	KKS1/KKS2 — standard variance categories
Period-end costing run	ICOST batch	CKMLCP — actual costing run
Plan costing	ICOST plan model	SAC → ACDOCP + CK40N plan cost estimates

 **Critical design decision:** ICOST cost components must be mapped to SAP cost component structure before cutover. Any mismatch between ICOST cost categories and SAP cost components will cause COGS split errors from day one. This mapping should be completed and validated in the Explore phase.

(Product Cost Controlling with SAP.pdf — product-costing)

4. End-to-End Architecture Overview



5. Material Ledger — What Changes in S/4HANA 2023

In S/4HANA, Material Ledger is **mandatory** for all plants. The data model has been significantly restructured.

5.1 Data Model Mapping (ECC → S/4HANA)

ECC Table	S/4HANA Replacement	Purpose
MLHD, MLIT, MLPP, MLPPF, MLCR, MLCRF, CKMLPP, CKMLCR	MLDOC and MLDOC_EXTRACT	ML transactional data + settlement
MLKEPH, CKMLKEPH	MLDOCCCS and MLDOCCCS_EXTRACT	Cost component split in actual costing
CKMLMV011	MLRUNLIST	Material/activity type status from costing run
MBEW, EBEW, OBEW, QBEW	ACDOCA (Universal Journal)	Inventory valuation

(Material Ledger in SAP S4HANA Functionality and Configuration.pdf — product-costing)

5.2 Key Simplification: Four-Step Close → Single Settlement

The four separate ECC closing steps — Single-Level Price Determination, Multilevel Price Determination, Revaluation of Consumption, WIP Revaluation — are **merged into one step called Settlement** in S/4HANA. Plan your cutover runbook around the new single-step model, not the ECC four-step model.

(SIMPL_OP2022.pdf — product-costing)

6. Material Extension — Per-Plant Requirements

Every material that will be valued in ML must have the correct master data configuration before CKMSTART runs. Missing or misconfigured materials at startup will not be initialized and will require remediation post-go-live, which is complex.

6.1 Material Master Settings Required

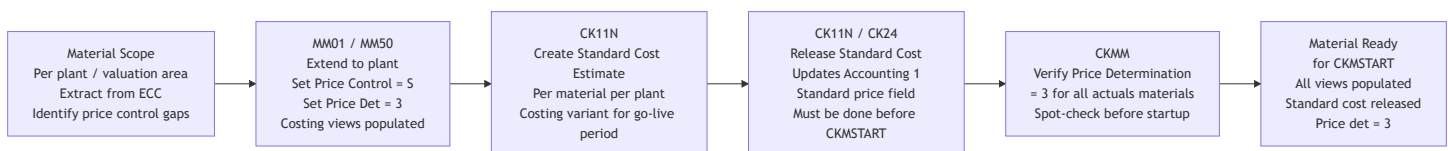
View	Field	Required Value	Notes
Costing 1	Price control	S (Standard)	ML actual costing uses standard as base — MAP materials not selected in

View	Field	Required Value	Notes
			actual costing run
Costing 1	Price determination	3	Enables actual costing; 2 = no ML actual costing
Costing 1	Costing lot size	Populated	Must match standard cost estimate
MRP 2	Plant-specific material status	Active	No restrictions that block goods movements
Accounting 1	Moving price	Populated	Starting price for initialization
Accounting 1	Standard price	Populated from CK11N release	Must be released before CKMSTART

⚠ **Critical:** Materials with price determination **2** and price control S (standard price without ML actual costing) are **not selected** in the actual costing run (CKMLCP). If ICOST previously valued materials using MAP logic, a deliberate decision is required on whether to migrate them to price determination 3 before go-live. Changing price determination from 3 to 2 post-go-live (via CKMM) causes all ML documents for that material to be excluded from closing — a significant audit risk.

(SAP-Material-LedgerOBYC.pdf — product-costing)

6.2 Per-Plant Extension Sequence



6.3 Pre-Requisite Checks Before CKMSTART (Per Plant)

Check	Transaction	Pass Condition
All prior ML costing runs complete	CKMLCP	No runs in status 'open' or partial
No incomplete CKMLCPAVR runs	CKMLCPAVR	All alternative valuation runs closed

Check	Transaction	Pass Condition
Standard cost estimates released for go-live period	CK24	All relevant materials have released standard
Price determination = 3 for actuals-scope materials	CKMM / SE16 T001W.MGVUPD	Field = X per plant
ML Type assigned to valuation area	OMX3	All plants in same company code → same ML type
ML currency types explicitly defined	OMX2	No default type "0000" — must be explicit in S/4HANA
PO history data archived / cleaned	EKBE, EKBZ	Reduce CKMSTART runtime
Manufacturing orders closed	CO02	All orders settled before startup

7. Material Ledger Cutover — The Core Cycle

7.1 The Fundamental Constraint

The ML startup program (**CKMSTART**) is the gate. **No goods movements, no new material masters, and no production order transactions** are permitted between ML activation in Customizing (OMX1) and the completion of CKMSTART. This window must be protected absolutely during the cutover weekend.

(Actual Costing with the SAP Material Ledger — product-costing)

7.2 What CKMSTART Does Per Plant

When run, CKMSTART:

1. Creates ML master data (CKMLHD, CKMLPR) for all existing materials in the plant
2. Translates all material prices and stock values into the ML currencies defined in OMX2/OMX3
3. Translates existing PO histories into the second and third ML currencies
4. Initializes cost component split data for existing inventory — sourced from the standard cost estimate; if no cost component split exists on the standard cost, the total is summarized into the

material consumption component

(Actual Costing with the SAP Material Ledger — product-costing)

7.3 Upload Cycle: Timing and Sequence

The ML upload cycle is period-boundary-driven. SAP's strong recommendation is to activate at the **beginning of a period** because ML begins updating the actual quantity structure from the moment of activation — any transactions before CKMSTART completion that are not captured will create gaps in the multilevel costing quantity structure.

Pre-Requisites Complete
OMX2/OMX3 configured
ML Type assigned to
valuation area
All prior costing runs closed
No open CKMLCP /
CKMLCPAVR



OMX1
Activate ML in Customizing
Per valuation area / plant
⚠ Goods movements now
locked



CK11N / CK40N
Create + Release
Standard Cost Estimates
For go-live period
Must exist BEFORE
CKMSTART



Extend Materials to ML
Material Master
MRP/Costing views
Price control = S
Price determination = 3
Per plant scope



CKMSTART
ML Startup Program
Run per plant batch

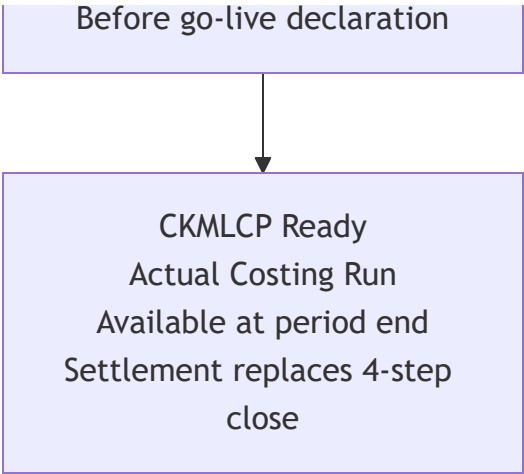
Per plant basis
Translates stock values +
PO history
Initializes cost component
split

LTMC / MB1C
Upload Initial Stock
Balances
Movement type 561
At standard cost
After CKMSTART – not
before

ML Active + Initialized
Goods movements re-
enabled
ML documents begin
posting
Actual quantity structure
building

CKM3N
Material Price Analysis
Validate: opening stock
value
= FI balance sheet balance

Finance Sign-Off
ML balance = GL balance
Per plant, per material
type

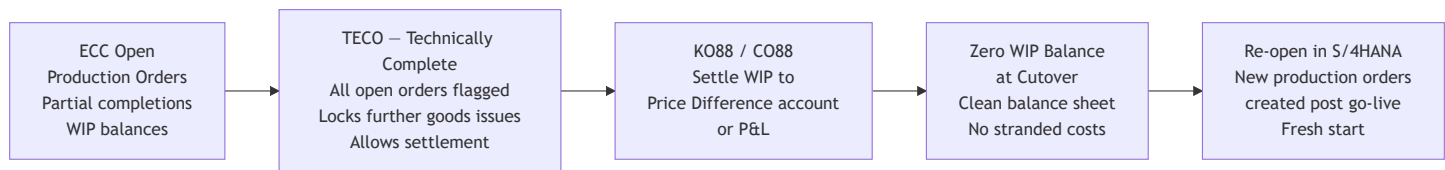


8. WIP Balances — Design Decision

This is the most complex element of the cutover given open production orders mid-flight. Three design options are evaluated below.

8.1 Option 1: Settle All WIP Before Cutover (Recommended)

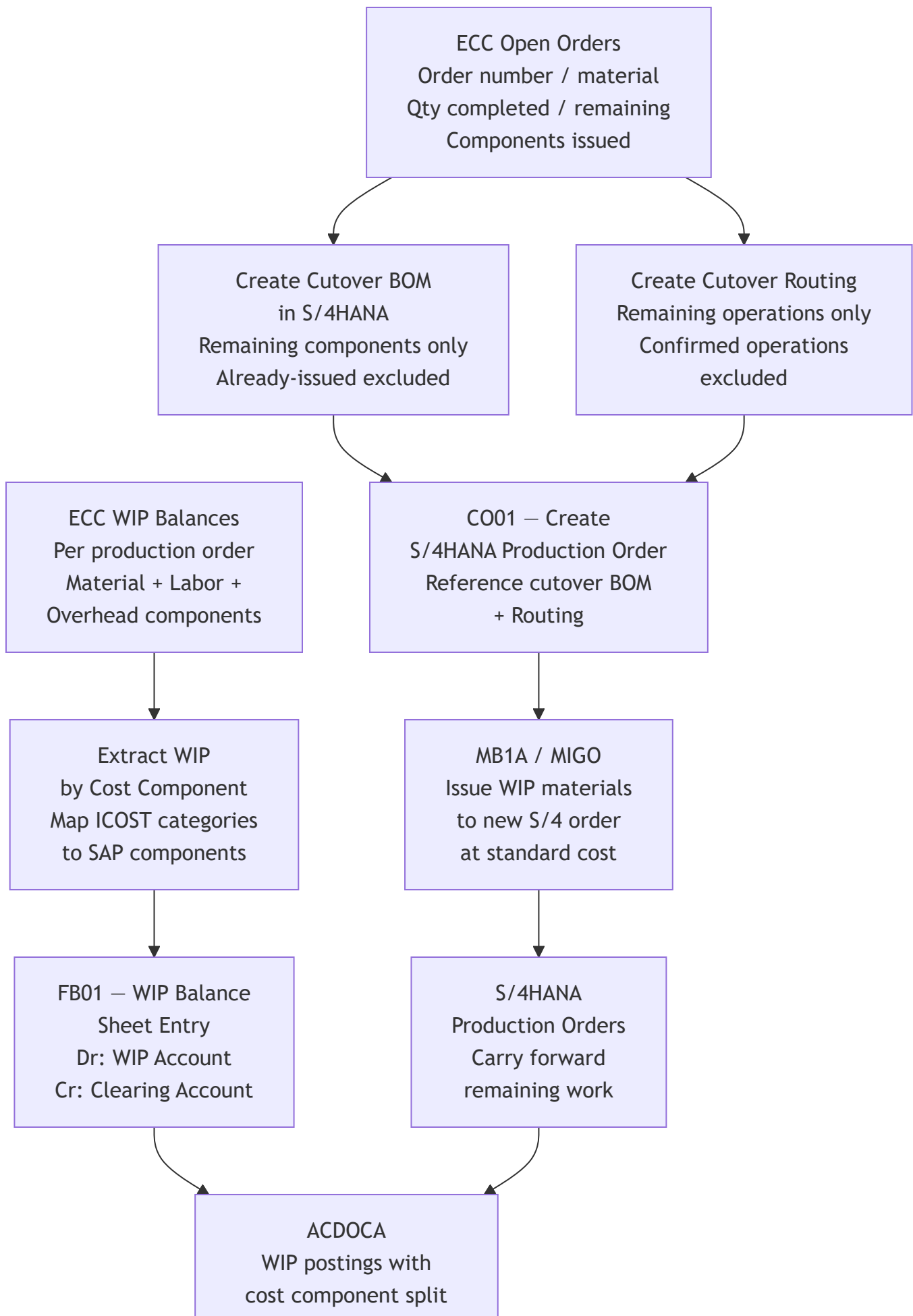
Thesis: Force all open production orders to a settleable state in ECC before go-live. Zero WIP at cutover.

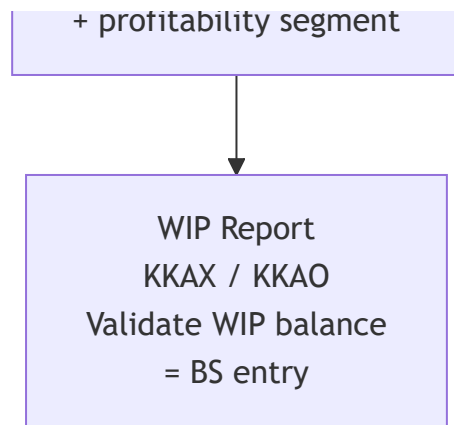


	Pros	Cons
1	Cleanest cutover — no WIP complexity in S/4HANA opening	Operationally disruptive — production must pause before go-live
2	FI = CO reconciliation is straightforward	Settlement of partially complete orders distorts period P&L
3	No special cutover BOM/routing required	May not be feasible for long-cycle manufacturing

8.2 Option 2: Load WIP as Balance Sheet Opening Entry (Pragmatic)

Thesis: Load WIP balances to the balance sheet via FB01, create shell production orders in S/4HANA, and continue operations post go-live.





Cutover BOM / Routing Design

Element	ECC Source	S/4HANA Cutover Design
BOM	Full production BOM	Remaining BOM — exclude already-issued components
Routing	Full routing with all operations	Remaining routing — confirmed operations marked done
Production Order	Open ECC order with WIP	New S/4 order referencing cutover BOM + routing
WIP Balance	Per ICOST cost components	FB01 BS entry Dr WIP / Cr Cutover Clearing
Goods Issue	Already issued in ECC	Issue remaining components only post go-live
Confirmation	Partially confirmed	Confirm remaining operations as executed

WIP GL Account Mapping

Dr/Cr	G/L Account	Description	Amount Basis
Dr	13100000	WIP — Raw Material Component	ICOST material cost
Dr	13200000	WIP — Labor	ICOST labor cost
Dr	13300000	WIP — Fixed Overhead	ICOST fixed OH
Dr	13400000	WIP — Variable Overhead	ICOST variable OH
Cr	19999999	Cutover Clearing Account	Total WIP balance

The clearing account is reversed post go-live once the production order in S/4HANA absorbs the costs through normal goods issue and confirmation postings.

(SAP Activate methodology for S4HANA cFIN.xlsx — product-costing)

8.3 Option 3: Direct Transfer via Migration Cockpit (Complex)

Thesis: Use LTMC to migrate open production orders directly with their WIP balances intact.

Aspect	Detail
LTMC Object	Production Order with components and operations
WIP handling	WIP balance migrated as order actual costs
ML impact	ML must be initialized before order migration
Risk	High — order status management complex across systems
Recommended?	Only for simple, short-cycle manufacturing

8.4 Verdict

Option 2 (Balance Sheet entry + Cutover BOM/Routing) is the most pragmatic and auditable approach for complex manufacturing with ICOST legacy costs.

9. Master Cutover Sequence

9.1 Pre-Cutover (Weeks Before Go-Live)

Step	Action	Owner	Transaction	Dependency
P1	Map ICOST cost components to SAP cost component structure	Costing team	OKTZ	None — must be first
P2	Activate Material Ledger per plant	BASIS/FI	OMX1 / SPRO	After ML config
P3	Configure COGS split accounts	FI	SPRO — OBYC	After GL master setup

Step	Action	Owner	Transaction	Dependency
P4	Configure OKB9 default account assignments	CO	OKB9	After GL + CC master
P5	Configure CO-PA characteristic derivation	CO-PA	KEA5 / KEA6	After segment design
P6	Configure assessment cycles	CO	KSU1	After CC hierarchy
P7	Create and validate cutover BOMs	PP	CS01	After BOM design
P8	Create and validate cutover routings	PP	CA01	After routing design
P9	Extract WIP balances from ECC by cost component	Finance	SE16 / COEP	After ICOST mapping
P10	Freeze ECC production orders	PP	CO02 — TECO partial	Before cutover weekend

9.2 Cutover Weekend — Day 1: Foundation

Step	Action	Transaction	Validates Against
D1-1	Lock ECC users	SU01 mass lock	All ECC access stopped
D1-2	Run ECC final period close	MMPV + F.16	ECC balance sheet final
D1-3	Extract final ECC trial balance	F.01	Reconciliation baseline

End of Document